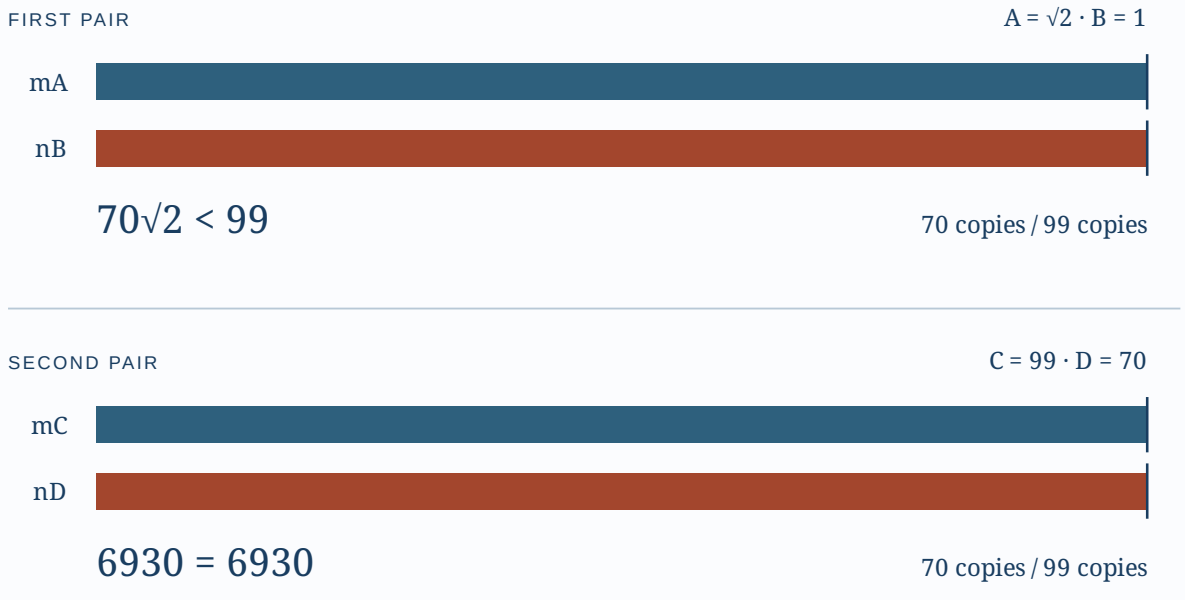
EUCLID V.5 / RETAINED EXPERIMENT

Thousands agree. One can disagree.

√2 : 1 and 99 : 70. Multipliers m = 70, n = 99.

Euclid V definitions 3–7, Heiberg’s 1884 Greek edition and nineteenth-century Latin translation, pp.2–5. English, examples, search, exact symbolic arithmetic and proof explanations are new teaching work. The square diagonal and comparison fractions are not examples printed in these definitions. A finite search can supply a counterexample to equality; agreement throughout that search is not a proof for every multiplier.

The selected equimultiples



First comparison

$mA < nB$

Second comparison

$mC = nD$

The endpoint microscope

×2,048 · automatic scale



Position shows (mA − nB)/nB or (mC − nD)/nD, with the same scale factor. Labels give exact offsets in length units. A dot locates the endpoint; its diameter is not a length measurement.

A counterexample to equal ratios: the first comparison gives “less than”; the second gives “equal”. This is not itself the strict-excess/non-excess pattern of definition 7. It still refutes the equality condition in definition 5.

Each pair uses its own drawing scale. Comparisons are within a pair, not between the two panels. Narrow gaps may be smaller than a pixel. More than 32 copies are shown as a continuous band rather than individual ticks.

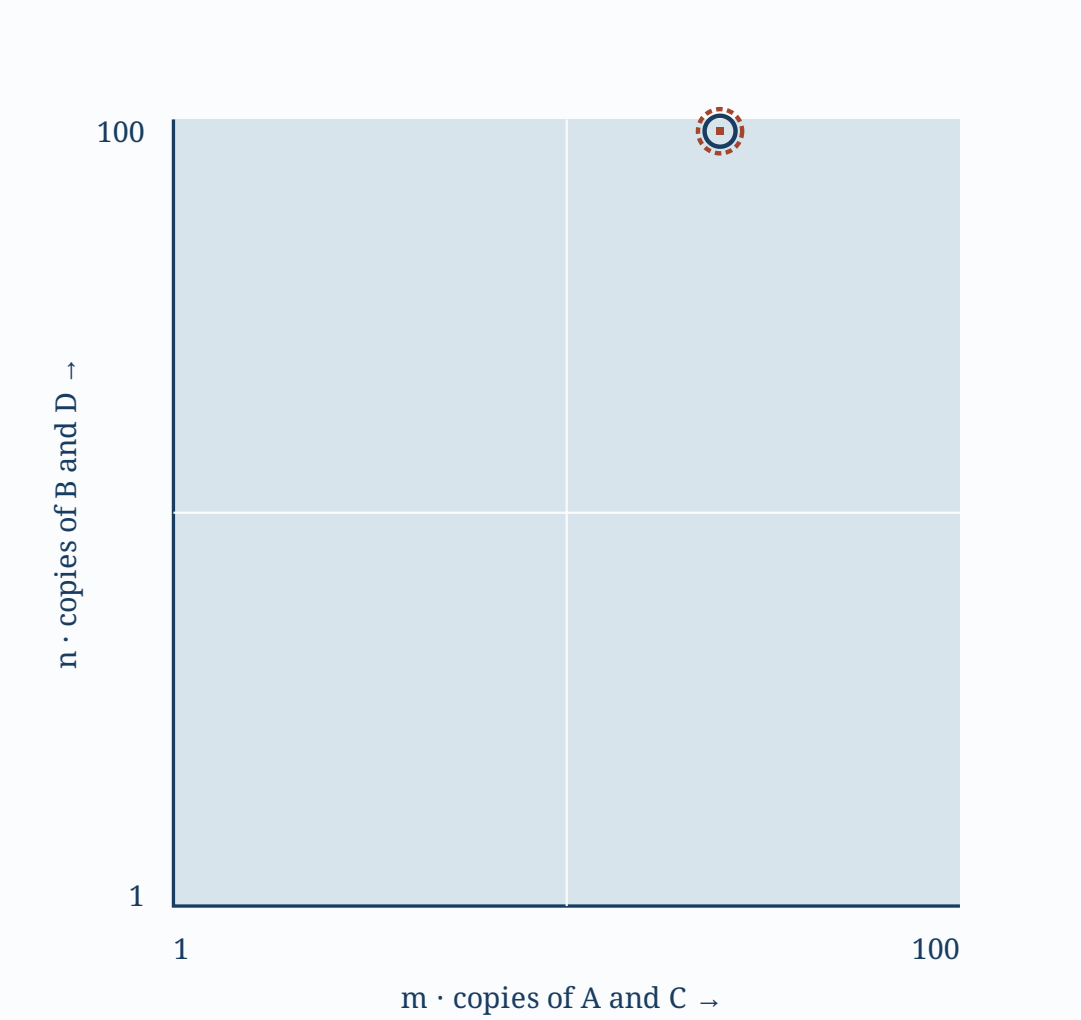
First pair. Both lengths are positive, so squaring preserves their order. $2 \times 70^2 = 9800$; $99^2 = 9801$.

Second pair. $70 \times 99 = 6930$; $99 \times 70 = 6930$.

The relation signs differ at these multipliers. Every sign uses integer arithmetic. The bars, microscope positions and decimal aids are approximate.

The finite search square

All positive integers m and n from 1 through 100: 10,000 exact comparisons, 1 disagreement. First in m-then-n order: m=70, n=99.



Blue denotes agreeing comparisons; rust denotes disagreements. The dashed ring locates the first disagreement; the solid ring locates the selected comparison if inside the square. Rings are locators, not extra cells. The search is a calculated set, not a record of comparisons individually performed by a learner.

Why no fraction can finish the job.

For this fraction, compare $p^2 = 9801$ with $2q^2 = 9800$. Their difference is 1. This integer calculation says which modern numerical ratio is larger; it is separate from a found equimultiple witness.

More generally, suppose $\sqrt{2} = p/q$ in lowest positive whole-number terms. Then $p^2 = 2q^2$, so p is even. Write $p = 2k$; then $q^2 = 2k^2$, so q is also even. Both terms would share a factor 2, contradicting lowest terms. No rational fraction equals $\sqrt{2}$.

This is a modern parity proof supplied for the example. It is not part of the definitions on these printed pages. A bounded search that finds no witness does not weaken the argument, and does not prove equality.

Read the condition

V · definition 3

1. A relation of magnitudes

GREEK · PRINTED P.2

Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἡ κατὰ πηλικότητά ποια σχέσις.

HEIBERG'S LATIN · PRINTED P.3

Ratio est duarum eiusdem generis magnitudinum secundum quantitatem quaelibet habitudo.

A ratio is a certain relation in size between two magnitudes of the same kind.

Here λόγος means ratio. ὁμογενῶν is genitive plural, agreeing with μεγεθῶν: 'of the same kind'. We compare lengths with lengths. A square's diagonal and side can have a ratio even when no length measures them both a whole number of times.

λόγος / ratio

ratio; the word has other meanings in other contexts.

μεγεθῶν / magnitudinum

of magnitudes; genitive plural, not a word for decimal numbers.

ὁμογενῶν / eiusdem generis

of the same kind. Both members of each ratio here are lengths.

V · definition 4

2. Able to exceed one another

GREEK · PRINTED P.2

Λόγον ἔχειν πρὸς ἄλληλα μεγέθη λέγεται, ἃ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέχειν.

HEIBERG'S LATIN · PRINTED P.3

Rationem inter se habere magnitudines dicuntur, quae multiplicatae altera alteram superare possunt.

Magnitudes are said to have a ratio to one another when, on being multiplied, they can exceed one another.

The condition concerns possible multiples, not the particular copies currently displayed. All positive lengths on this bench satisfy it. The horn-angle workshop shows why that condition deserves attention for other kinds of magnitude.

πολλαπλασιαζόμενα / multiplicatae

being multiplied, or taken repeatedly.

ὑπερέχειν / superare

to exceed. Equality does not satisfy this word.

V · definition 5, complete

3. The complete test for the same ratio

GREEK · PRINTED P.2

Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρὸς δεύτερον καὶ τρίτον πρὸς τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ' ὁποιονοῦν πολλαπλασιασμὸν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέχῃ ἢ ἅμα ἴσα ᾖ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.

HEIBERG'S LATIN · PRINTED P.3

In eadem ratione magnitudines esse dicuntur prima ad secundam et tertia ad quartam, si primae et tertiae aeque multiplices secundae et quartae aeque multiplices aut simul superant aut simul aequales sunt aut simul minores sunt suo ordine sumptae.

Magnitudes are said to have the same ratio, the first to the second and the third to the fourth, when equal-times multiples of the first and third, compared respectively with equal-times multiples of the second and

fourth, for whatever multiplication, either both exceed, both are equal, or both fall short.

‘Equal-times’ pairs the first with the third, and separately the second with the fourth. It does not require all four to have the same multiplier. Heiberg’s p.5 footnote makes that distinction explicit. The modern letters m and n name those two choices. The universal condition ranges over every pair of positive integers.

ἰσάκις / aeque

an equal number of times, within the specified pair.

ἅμα / simul

together, alike: the corresponding comparisons must have the same outcome.

κατάλληλα / suo ordine

in corresponding order: first with second; third with fourth.

V · definition 5, selected clause

4. Whatever multiplication

GREEK · PRINTED P.2

καθ’ ὁποιονοῦν πολλαπλασιασμὸν

HEIBERG’S LATIN · PRINTED P.3

primae et tertiae aeque multiplices secundae et quartae aeque multiplices

For whatever multiplication: choose any positive whole number m for the first and third, and independently any positive whole number n for the second and fourth.

The Latin displayed here is the surrounding multiplier clause, not a word-for-word counterpart of the short Greek phrase. Heiberg supplies no separate phrase for ‘whatever multiplication’ in his Latin sentence. His p.3 footnote gives a symbolic explanation. That printed algebra is nineteenth-century commentary, not ancient notation. No finite upper limit occurs in the Greek condition.

ὁποιονοῦν / no separate phrase here

of whatever kind; the indefinite force is essential to the test.

πολλαπλασιασμόν / multiplices (in context)

multiplication, taking a multiple; accusative after κατά.

V · definition 5, selected clause

5. Three outcomes, all retained

GREEK · PRINTED P.2

ἢ ἅμα ὑπερέχῃ ἢ ἅμα ἴσα ᾖ ἢ ἅμα ἐλλείπῃ

HEIBERG'S LATIN · PRINTED P.3

aut simul superant aut simul aequales sunt aut simul minores sunt

Either both exceed, or both are equal, or both fall short.

A less-than comparison paired with equality is already a disagreement. Do not throw away the equality case because a drawing's endpoints look coincident. The modern exact check compares integer squares for m√2 and n, and integer products for mp and nq.

ἴσα / aequales

equal; one of three separate comparison outcomes.

ἐλλείπῃ / minores sunt

falls short; the Latin expresses this as 'are smaller'.

V · definition 6

6. Give the relation its name

GREEK · PRINTED P.2

Τὰ δὲ τὸν αὐτὸν ἔχοντα λόγον μεγέθη ἀνάλογον καλείσθω.

HEIBERG'S LATIN · PRINTED P.3

Magnitudines autem eandem rationem habentes proportionales uocentur.

Let magnitudes having the same ratio be called proportional.

The name follows the previous definition's condition. It does not replace that condition with 'the displayed decimal values look the same'. The threefold pair on this bench has the same ratio by a general scaling argument.

ἀνάλογον / proportionales

proportional, having the same ratio.

καλείσθω / uocentur

let them be called; a naming instruction.

V · definition 7, complete

7. Greater, with a witness

GREEK · PRINTED P.4

Ὅταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέχῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέχῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρὸς τὸ δεύτερον μείζονα λόγον ἔχειν λέγεται, ἤπερ τὸ τρίτον πρὸς τὸ τέταρτον.

HEIBERG'S LATIN · PRINTED P.5

Sin ex aeque multiplicibus primae multiplex multiplicem secundae superat, tertiae autem multiplex multiplicem quartae non superat, tum prima ad secundam maiorem rationem habere dicitur quam tertia ad quartam.

When, among the equal-times multiples, the multiple of the first exceeds that of the second, while the multiple of the third does not exceed that of the fourth, the first is said to have a greater ratio to the second than the third has to the fourth.

'Does not exceed' allows equality. A disagreement always refutes the condition for equal ratios, but a particular disagreement need not have definition 7's strict-excess/non-excess form. The bench distinguishes those two conclusions. If the second pair supplies the excess, exchange the two ratio pairs to apply the definition.

μὴ ὑπερέχῃ / non superat

does not exceed: less than or equal, not only strictly less.

μείζονα λόγον / maiorem rationem

a greater ratio. The witness determines which pair has it.

The complete printed evidence

J. L. Heiberg, *Euclidis opera omnia, Elementa*, volume II, Leipzig: B. G. Teubner, 1884. Title and printed pp.2–5, corresponding to PDF pp.7 and 28–31. University of Toronto, Gerstein Science Information Centre copy, digitized by Internet Archive. Public domain.

Whole pages are reduced without cropping or retouch. Line wrapping and end-of-line hyphenation are removed in the reading; Greek theta glyphs are normalized. Latin u/v spellings remain. Superscript note markers and apparatus omitted from selections remain in these pages. Definitions 3–7 are transcribed in full across reading units; two units revisit selected clauses of V.5. The Latin excerpt in “Whatever multiplication” is explicitly a surrounding clause rather than an invented phrase. Footnotes containing algebra on pp.3 and 5 belong to the modern editor. New English and language notes have not had independent philological peer review.

Complete original source PDF

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ELEMENTA.

EDIDIT ET LATINE INTERPRETATUS EST

I. L. HEIBERG,

DR. PHIL.

UOL. II.

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MDCCCLXXXIV.

ε′.

Ὅροι.

α′. *Μέρος ἐστὶ μέγεθος μεγέθους τὸ ἔλασσον τοῦ μείζονος, ὅταν καταμετρῇ τὸ μεῖζον.*

β′. *Πολλαπλάσιον δὲ τὸ μεῖζον τοῦ ἐλάττονος, ὅταν καταμετρῆται ὑπὸ τοῦ ἐλάττονος.*

γ′. *Λόγος ἐστὶ δύο μεγεθῶν ὁμογενῶν ἡ κατὰ πηλικότητά ποια σχέσις.*

δ′. *Λόγον ἔχειν πρὸς ἄλληλα μεγέθη λέγεται, ἃ δύναται πολλαπλασιαζόμενα ἀλλήλων ὑπερέχειν.*

ε′. *Ἐν τῷ αὐτῷ λόγῳ μεγέθη λέγεται εἶναι πρῶτον πρὸς δεύτερον καὶ τρίτον πρὸς τέταρτον, ὅταν τὰ τοῦ πρώτου καὶ τρίτου ἰσάκις πολλαπλάσια τῶν τοῦ δευτέρου καὶ τετάρτου ἰσάκις πολλαπλασίων καθ' ὁποιονοῦν πολλαπλασιασμὸν ἑκάτερον ἑκατέρου ἢ ἅμα ὑπερέχῃ ἢ ἅμα ἴσα ᾖ ἢ ἅμα ἐλλείπῃ ληφθέντα κατάλληλα.*

ς′. *Τὰ δὲ τὸν αὐτὸν ἔχοντα λόγον μεγέθη ἀνάλογον καλείσθω.*

Def. 1. Hero def. 120, 1. Barlaam logist. I def. 1. 2. Hero def. 121. Barlaam I def. 2. 3. Hero def. 127. Psellus p. 8. 4. Hero def. 123, 1. 5. Hero def. 124. 6. Hero def. 124.

---

1. ὅροι] om. PBFp. numeros om. codd. omnes. 2. ἔλαττον Hero. 4. ἐλάσσονος V, ut lin. 5. 7. ποια] P, Hero; πρὸς ἄλληλά ποια Theon (BFV p), Campanus. Post σχέσις add. ἀναλογία δὲ ἡ τῶν λόγων ταυτότης Bp, Campanus; mg. m. 2 P V; mg. bis m. 1 et m. 2 F; om. Hero. 8. ἔχειν]

Liber V.

Definitiones.

1. Pars est minor magnitudo maioris, si maiorem metitur.

2. Multiplex autem maior est minoris, si minor eam metitur.

3. Ratio est duarum eiusdem generis magnitudinum secundum quantitatem quaelibet habitudo.

4. Rationem inter se habere magnitudines dicuntur, quae multiplicatae altera alteram superare possunt.

5. In eadem ratione magnitudines esse dicuntur prima ad secundam et tertia ad quartam, si primae et tertiae aeque multiplices secundae et quartae aeque multiplices aut simul superant aut simul aequales sunt aut simul minores sunt suo ordine[1]) sumptae.

6. Magnitudines autem eandem rationem habentes proportionales uocentur.

---

1) Hoc est: ita ut coniungantur prima secundae, tertia quartae et respondeat loco et ordine prima tertiae, secunda quartae. itaque si $Ma \gtreqless Nb$ et simul $Mc \gtreqless Nd$, erit $a : b = c : d$. cfr. Hankel: Zur Gesch. der Mathemat. p. 390.

ν supra m. 1 P. 9. *ὑπερέχειν*] *-ειν* in ras. V. 14. *πολλαπλασιασμῶν* P, corr. m. 1. 15. *ὑπερέχει* B. *ἥ*] supra m. 1 F. *ἐλλείπει* B. *ληφθέντα*] *-η-* e corr. m. 2 V. Deff. 6—7 permutauit P; ut nos BFVp, Campanus; ex Herone nihil concludi potest, cum etiam def. 8—9 ante def. 7 habeat. 17. *ἔχοντα λόγον μεγέθη*] *λόγον ἔχοντα μεγέθη* F; *ἔχοντα μεγέθη λόγον* V. *ἀνάλογον*] *λόγον ἀνάλογον* post ras. 7 litt. in mg. transit m. 2 F.

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ξ΄. Ὅταν δὲ τῶν ἰσάκις πολλαπλασίων τὸ μὲν τοῦ πρώτου πολλαπλάσιον ὑπερέχῃ τοῦ τοῦ δευτέρου πολλαπλασίου, τὸ δὲ τοῦ τρίτου πολλαπλάσιον μὴ ὑπερέχῃ τοῦ τοῦ τετάρτου πολλαπλασίου, τότε τὸ πρῶτον πρὸς τὸ δεύτερον μείζονα λόγον ἔχειν λέγεται, ἤπερ τὸ τρίτον πρὸς τὸ τέταρτον.

η΄. Ἀναλογία δὲ ἐν τρισὶν ὅροις ἐλαχίστη ἐστίν.

θ΄. Ὅταν δὲ τρία μεγέθη ἀνάλογον ᾖ, τὸ πρῶτον πρὸς τὸ τρίτον διπλασίονα λόγον ἔχειν λέγεται ἤπερ πρὸς τὸ δεύτερον.

ι΄. Ὅταν δὲ τέσσαρα μεγέθη ἀνάλογον ᾖ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔχειν λέγεται ἤπερ πρὸς τὸ δεύτερον, καὶ ἀεὶ ἑξῆς ὁμοίως, ὡς ἂν ἡ ἀναλογία ὑπάρχῃ.

ια΄. Ὁμόλογα μεγέθη λέγεται τὰ μὲν ἡγούμενα τοῖς ἡγουμένοις τὰ δὲ ἑπόμενα τοῖς ἑπομένοις.

ιβ΄. Ἐναλλὰξ λόγος ἐστὶ λῆψις τοῦ ἡγουμένου πρὸς τὸ ἡγούμενον καὶ τοῦ ἑπομένου πρὸς τὸ ἑπόμενον.

ιγ΄. Ἀνάπαλιν λόγος ἐστὶ λῆψις τοῦ ἑπομένου ὡς ἡγουμένου πρὸς τὸ ἡγούμενον ὡς ἑπόμενον.

ιδ΄. Σύνθεσις λόγου ἐστὶ λῆψις τοῦ ἡγουμένου μετὰ τοῦ ἑπομένου ὡς ἑνὸς πρὸς αὐτὸ τὸ ἑπόμενον.

---

7. Hero def. 125, 5. 8. Hero def. 124. 9. Hero def. 125, 1. 11. Hero def. 126. 12. Hero def. 127, 6. 13. Hero def. 127, 2. 14. Hero def. 127, 3.

---

2. *ὑπερέχει* P, sed corr. m. 1. 4. *τὸ*] supra m. 1 V. Post def. 7 seq. *ἀναλογία δέ ἐστιν ἡ τῶν λόγων ὁμοιότης* Fp et V (del. punctis, sed puncta erasa); om. PB, Hero, Campanus. 7. *τρισίν*] *-ισ-* in ras. m. 2 V. *ἐλαχίστοις* V. Def. 10 om. Heron. 12. *τό*] om. P. *τριπλασίονα*] *τρι-* in ras. p. 13. *ἀεί*] *αἰεί* FV. *καὶ ἀεί* — 14: *ὑπάρχῃ*] om. Campanus. 13. *ὁμοίως*] P; *ἑνὶ πλείους* Theon (BFVp). 14. *ὡς*]

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7. Sin ex aeque multiplicibus[1]) primae multiplex multiplicem secundae superat, tertiae autem multiplex multiplicem quartae non superat, tum prima ad secundam maiorem rationem habere dicitur quam tertia ad quartam.

8. Proportio autem in tribus terminis consistens minima est.

9. Si tres magnitudines proportionales[2]) sunt, prima ad tertiam duplicatam rationem quam ad secundam habere dicitur.

10. Sin quattuor magnitudines proportionales[3]) sunt, prima ad quartam triplicatam rationem quam ad secundam habere dicitur, et eodem modo semper deinceps, qualiscunque data est proportio.

11. Respondentes magnitudines dicuntur praecedentes praecedentibus, sequentes sequentibus.

12. Permutata ratio est, ubi sumitur praecedens ad praecedentem et sequens ad sequentem.

13. Inuersa ratio est, ubi sumitur sequens praecedentis loco ad praecedentem sequentis loco.

14. Compositio rationis est, ubi sumitur praecedens cum sequenti pro una ad solam sequentem.

---

1) Non omnes aeque multiplices esse debent, sed primae et tertiae aeque multiplices, secundae et quartae, ut in def. 5.

2) Sc. deinceps (*κατὰ τὸ συνεχές*), h. e. si $a : b = b : c$, erit $a : c = a^2 : b^2$.

3) Sc. deinceps (*κατὰ τὸ συνεχές*); cfr. XI, 33. h. e. si $a : b = b : c = c : d$, erit $a : d = a^3 : b^3$.

ἕως FV, p m. rec. 15. *ἡγούμενα*] *ἡ-* e corr. m. 2 V. 16. *τὰ δὲ ἑπόμενα τοῖς*] m. 2 in ras. V. 19. *ἐστίν* F. 21. *ἐστίν* B. *τοῦ*] insert. m. 2 F.

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Reopen the experiment

```
{
  "format": "stoicheia-ratios",
  "version": 1,
  "state": {
    "kind": "fraction",
    "p": 99,
    "q": 70,
    "m": 70,
    "n": 99,
    "bound": 100,
    "reading": 6,
    "page": 5
  }
}
```

The CSV separates the selected comparison from inclusive n-intervals containing every disagreement for each m in the bounded search. It does not assert a learner action history. The JSON stores settings; the arithmetic is recomputed on import.

Reopen these settings in the workshop. This app link requires the corresponding workshop build. This retained reading itself needs no scripts or network subresources.


```
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